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FLAIR: VLM with Fine-grained Language-informed Image Representations

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Abstract

*CLIP has shown impressive results in aligning images and texts at scale. However, its ability to capture detailed visual features remains limited because CLIP matches images and texts at a global level. To address this issue, we propose FLAIR, **F**ine-grained **La**nguage-informed **I**mage **R**epresentations, an approach that utilizes long and detailed image descriptions to learn localized image embeddings. By sampling diverse sub-captions that describe fine-grained details about an image, we train our vision-language model to produce not only global embeddings but also text-specific image representations. Our model introduces text-conditioned attention pooling on top of local image tokens to produce fine-grained image representations that excel at retrieving detailed image content. We achieve state-of-the-art performance on both, existing multimodal retrieval benchmarks, as well as, our newly introduced fine-grained retrieval task which evaluates vision-language models' ability to retrieve partial image content. Furthermore, our experiments demonstrate the effectiveness of FLAIR trained on 30M image-text pairs in capturing fine-grained visual information, including zero-shot semantic segmentation, outperforming models trained on billions of pairs. Code is available at https://github.com/ExplainableML/flair.*

1. Introduction

By encoding images and texts into global embeddings, CLIP [41] achieves coarse-grained semantic understanding. However, it loses track of the local image details, e.g., CLIP is not able to perceive the difference between "background" and "frappucino", resulting in the inability to highlight the relevant regions specified in the text prompt, as illustrated in Fig. 1. Recently, it has been shown that CLIP models and other Vision Language Models (VLMs) often lack visual details [47, 48]. Thus, our goal is to improve the fine-grained visual understanding of CLIP models, which is essential for a wide range of downstream applications, such as image-text retrieval or semantic segmentation.

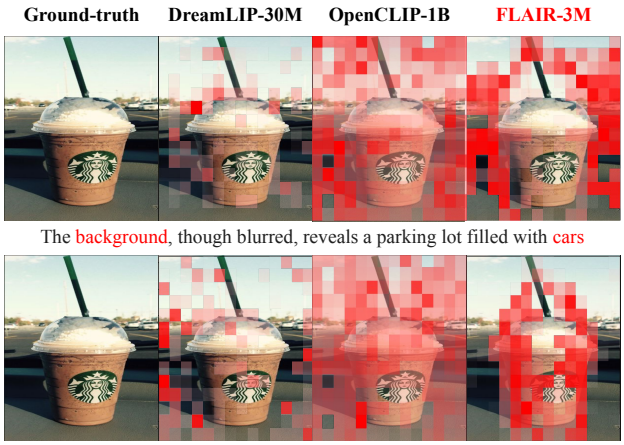


Figure 1. Visualization of the similarity scores between local image tokens and different text queries. While previous works [41, 60] lack fine-grained alignment, FLAIR matches text and image semantics at the token level.

Previous works [14, 60] propose to generate detailed descriptions for images to achieve more localized image-text alignment in CLIP models. However, these methods are restricted by the conventional learning mechanism of CLIP, since the detailed text descriptions enhance visual representations indirectly by matching them through the contrastive loss. Although DreamLIP [60] proposed to supervise local image tokens with textual information, we find that, without a careful selection of the negative pairs in the contrastive loss, the VLM does not learn to align the image tokens with semantically matching text, as illustrated in Fig. 1.

To address these issues, we propose FLAIR, to learn **F**ine-grained **La**nguage-informed **I**mage **R**epresentations, where image embeddings are generated by conditioning on a relevant text embedding for a more targeted alignment, instead of an indirect alignment through a global loss function. To obtain image descriptions with maximum semantic richness, our method leverages long-caption datasets generated by Multimodal Large Language Models (MLLMs).

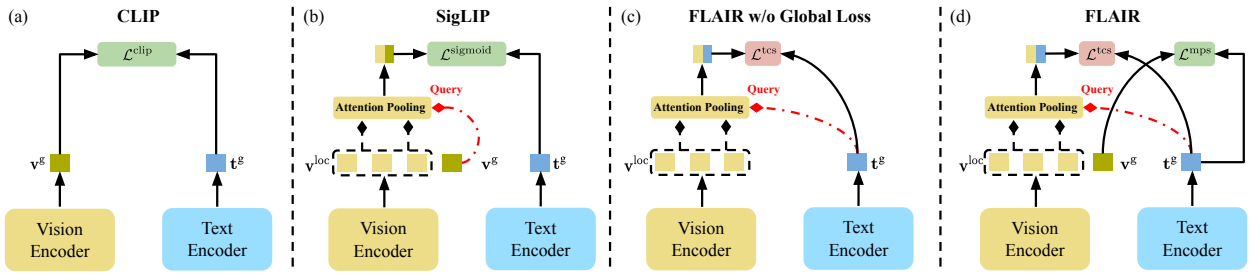


Figure 2. Comparison of text-conditioned attention pooling with previous methods. (a) Vanilla CLIP ($\mathcal{L}^{clip}$) aligns global image $\mathbf{v}^g$ and text $\mathbf{t}^g$ tokens. (b) SigLIP ($\mathcal{L}^{sigmoid}$) employs a global learnable image token $\mathbf{v}^g$ as query for a cross-attention layer to pool the local tokens $\mathbf{v}_{loc}$. (c) FLAIR ($\mathcal{L}^{tcs}$) employs text-conditioned attention pooling that leverages $\mathbf{t}^g$ as query, aggregating $\mathbf{v}^{loc}$ to capture language-informed visual features. (d) FLAIR ($\mathcal{L}^{tcs} + \mathcal{L}^{mps}$) adds an extra multi-positive global sigmoid loss to refine global-level image-text alignment.

These captions provide a rich source of information about specific objects or regions in the image. Given a long caption, we sample diverse sub-captions, some of which focus on local regions, while others describe the image globally. Considering that these captions describe the image to varying extents, we design our image encoder to produce text-conditioned image representations. To be specific, we introduce an attention pooling operation that uses the caption as a query to pool relevant image-token embeddings together.

As a result, FLAIR learns fine-grained image embeddings that demonstrate strong performance at retrieving fine-grained visual information. As shown in Figure 1, FLAIR can localize image regions relevant to the fine-grained textual description simply by computing the embedding similarity with respect to the individual image tokens. This is in contrast to previous methods that fail to capture local similarity. To analyze the text-image retrieval capabilities of our model, we consider three settings: standard (global) captions, long captions, and a newly proposed fine-grained retrieval setting, where the goal is to match short captions that describe a local region of the image. Our experimental evaluation on multimodal retrieval and zero-shot semantic segmentation demonstrates that FLAIR, trained on 30M image-text pairs with long synthetic captions, significantly outperforms previous vision-language models trained on billions of image-text pairs. While excelling at fine-grained tasks, FLAIR demonstrates comparable performance on global-level tasks, such as image classification, when trained on the same amount of data.

Our key contributions can be summarized as follows: 1) We propose FLAIR, a model architecture that employs text-conditioned attention pooling to produce fine-grained and localized image embeddings. 2) Building upon long synthetic captions, we introduce a diverse caption sampling strategy to obtain a rich set of positive and negative image-text pairs facilitating the learning of global and local multimodal relations. 3) Our experimental evaluation on fine-grained downstream tasks shows that FLAIR, trained on 30M samples, outperforms previous models by up to 10.8% R@1 on coarse-to-fine multimodal retrieval and by up to 11.2% R@1 on long retrieval tasks. Compare to CLIP models trained on billions of data, FLAIR achieves an average of 14.4% increase in mIoU on segmentation tasks.

2. Related Works

Vision-Language Pre-training. CLIP [41] and ALIGN [21] have scaled up vision-language pre-training datasets to 400M and 1B samples, using a contrastive loss to match global image tokens with global text tokens (Fig. 2 (a)). However, there is a growing demand for more fine-grained alignment between modalities [48]. Several approaches have been proposed to achieve this goal, including token-level alignment [55], hierarchical alignment from global to local [17], soft assignments allowing many-to-many mappings [18], and the use of intra-modal contrastive losses [27]. CoCa [56] utilized cross-attention to pool local image tokens and achieved more refined image-to-text alignment by additionally training with a captioning objective. Along with attention pooling to form the global image embeddings (Fig. 2 (b)), SigLIP [58] replaced the Softmax loss of vision-language pre-training with a Sigmoid-based loss. Concurrent to our work, Llip [26] have proposed an architecture that incorporates language information into learnable image tokens to form contextualized visual representations. However, Llip [26] lacks the pooling of local image tokens (Fig. 2 (c)) and, thus, it does not ensure a fine-grained alignment between modalities. In contrast, FLAIR leverages diverse and detailed captions with both local and global alignment (Fig. 2 (d)), outperforming previous approaches even when training on a significantly smaller dataset.

Text Augmentation. Several works [14, 51, 59, 60] proposed to improve visual-language alignment through text augmentation. Notably, LaCLIP [14] rewrited captions in large datasets with Large Language Models (LLMs), showing significant performance gains when training on synthetic captions. Similarly, large Vision-Language Models

(VLMs) have been exploited to create synthetic images and captions, augmenting existing datasets [20, 31, 54]. DreamLIP [60] re-captioned 30 million images from CC3M [45], CC12M [4] and YFCC15M [9] with detailed descriptions generated by pre-trained MLLMs. Employing these synthetic captions, several models have been trained to handle long texts going beyond the 77-token limit of CLIP. LongCLIP [59], LoTLIP [51], and TULIP [35] all leveraged synthetic captions to achieve this goal. Although trained on the same 30M re-captioned images as DreamLIP, FLAIR changes the image-text interaction by directly using text-conditioned attention pooling to aggregate the local image tokens and choosing informative negative pairs in the loss function. Notably, without modifying the text encoder, the diverse sampling strategy empowers FLAIR to surpass models specialized for the long caption retrieval task.

3. FLAIR: Fine-grained Language-informed Image Representations

In this section, we present the FLAIR architecture and methodology for language-image pre-training. We provide an overview of the main components of FLAIR in Fig. 3, including the sampling of diverse captions from long synthetic descriptions (Sec. 3.1), the text-conditioned attention pooling of image tokens (Sec. 3.2), and the local and global loss functions (Secs. 3.3 and 3.4).

3.1. Sampling Diverse Captions

Pre-training data for vision-language models is typically collected by scraping and filtering large amounts of web data, as performed by CC3M [45] or LAION [43, 44]. While a large amount of image-text pairs helps in discovering a comprehensive set of visual concepts, the text descriptions in these datasets often do not describe the image content in detail. As a result, it is not possible to extract fine-grained concepts in an image, such as scene composition, and small object features. To alleviate this issue, we employ image datasets that are synthetically re-captioned and contain a long and detailed description of each image [46, 60]. A single sentence of these long captions typically describes a particular image detail, e.g., one object, a feature of an object, the background, the image style, or context.

Using these captions, our goal is to align vision and language representations at the fine-grained level of individual caption sentences, while retaining global image understanding. We devise a sampling strategy to cover both local and global captions, and learn their similarity with adaptively pooled image features through a contrastive loss. Specifically, when constructing a batch of B images from the augmented dataset, we sample K sub-captions from a caption T_i belonging to an image I_i. Each sub-caption consists of $s \in \{1, \dots, S\}$ sentences that are either randomly sampled (i.e., independently sampled and concatenated), or extracted as a consecutive sequence of sentences. As a result, a batch contains B images and $B \times K$ texts, where each image is associated with K matching captions. At each iteration, we randomly choose the number of sentences s for every sub-caption, where a lower s result in a more localized caption, while more sentences (a higher s) describe multiple parts of the image, resulting in global descriptions. We provide examples on the original long caption and our sampled captions in Sec. B in the supplementary.

3.2. Text-conditioned Attention Pooling

Having access to a diverse set of captions, some describing local regions of an image and others explaining the global content, motivates creating a model architecture that is capable of adapting to both scenarios. Naively applying a contrastive loss between a global image embedding and the individual text embeddings would collapse the carefully separated information content of our K captions into an averaged image representation.

Instead, we propose to contextualize the image representations with the individual captions, producing a unique image representation for every image-text pair. We start with the VLM architecture as proposed by Radford et al. [41], which uses two independent transformer encoders f_{img} and f_{txt} to project the tokenized image and text samples into per-token embeddings and global embeddings (i.e., $f_{\text{img}}(I) = [\{\mathbf{v}^{(p)}\}_{p=1}^{n}, \mathbf{v}^{\text{g}}]$ and $f_{\text{txt}}(T) = [\{\mathbf{t}^{(p)}\}_{p=1}^{m}, \mathbf{t}^{\text{g}}]$), where n is the number of image tokens and m the number of text tokens. For simplicity, we refer to the local image tokens $\{\mathbf{v}^{(p)}\}_{p=1}^{n}$ as $\mathbf{v}^{\text{loc}} \in \mathbb{R}^{n \times d}$, where d denotes the embedding dimension.

To effectively contextualize the image representation with semantics from the sampled captions, we introduce an attention pooling layer f_{AttnPool}, that produces a text-conditioned image representation $\mathbf{v}^{\text{tc}}$ from the local image patch embeddings and the global text embedding. We define $\mathbf{v}^{\text{tc}} = f_{\text{AttnPool}}(\mathbf{t}^{\text{g}}, \mathbf{v}^{\text{loc}})$ as follows:

$$f_{\text{AttnPool}}(\mathbf{t}^{\text{g}}, \mathbf{v}^{\text{loc}}) = \text{softmax}\left(\frac{\mathbf{t}^{\text{g}} W_{\text{q}} (\mathbf{v}^{\text{loc}} W_{\text{k}})^{\text{T}}}{\sqrt{d}}\right) \mathbf{v}^{\text{loc}} W_{\text{v}} \tag{1}$$

where W_q, W_k, W_v are the query, key, and value weight matrices. In other words, we use the global text embeddings of a caption as a query to pool the local image embeddings creating a text-conditioned image representation $\mathbf{v}^{\text{tc}}$. In practice, we use a multi-head attention layer. We append an empty token (zero vector) to $\mathbf{v}^{\text{loc}}$ to allow $\mathbf{t}^{\text{g}}$ to attend to the empty token when $\mathbf{t}^{\text{g}}$ and $\mathbf{v}^{\text{loc}}$ are not semantically related.

Choosing Negative Pairs. With text-conditioned attention pooling, FLAIR produces a different image representation for every image-text pair. However, to learn semantically rich and nuanced image representations, we must carefully define the positive and negative pairs for vision-language pre-training. To simplify the notations, we assume a single

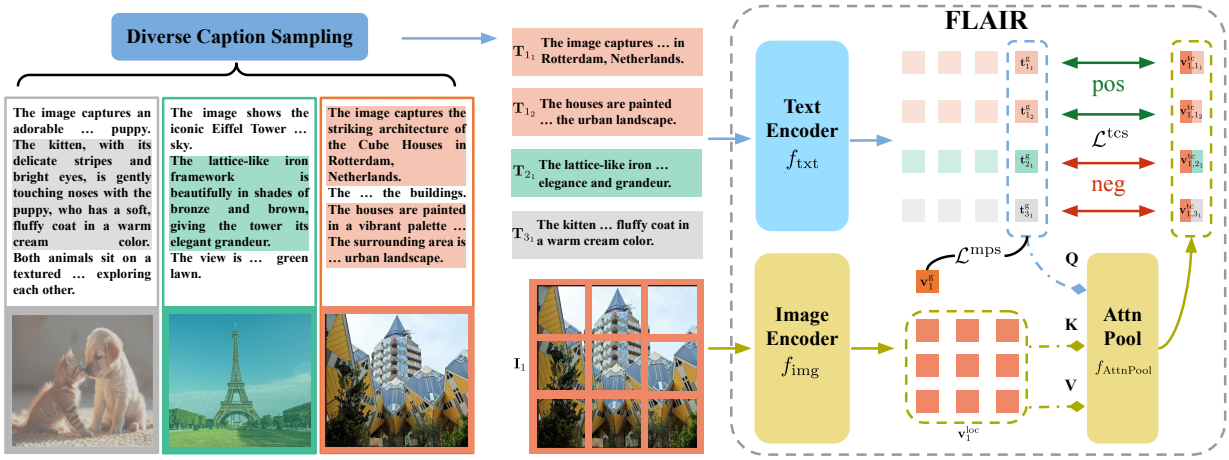


Figure 3. Overview of FLAIR; We sample diverse positive and negative captions $\{T_{1_1}...T_{3_1}\}$ for an image I_1. f_{txt} and f_{img} then produce the global text tokens $\{\mathbf{t}^{\text{g}}_{1_1}...\mathbf{t}^{\text{g}}_{3_1}\}$, the global image token $\mathbf{v}^{\text{g}}_1$, and local image tokens $\mathbf{v}^{\text{loc}}_1$. Conditioned on $\{\mathbf{t}^{\text{g}}_{1_1}...\mathbf{t}^{\text{g}}_{3_1}\}$, $f_{\text{AttnPool}(.)}$ generates fine-grained text-conditioned image representations $\{\mathbf{v}^{\text{tc}}_{1_1}...\mathbf{v}^{\text{tc}}_{3_1}\}$. The text-conditioned sigmoid loss $\mathcal{L}^{\text{tcs}}$ aligns $\{\mathbf{t}^{\text{g}}_{1_1}...\mathbf{t}^{\text{g}}_{3_1}\}$ with $\{\mathbf{v}^{\text{tc}}_{1_1}...\mathbf{v}^{\text{tc}}_{3_1}\}$ contrastively, while the multi-positive sigmoid loss $\mathcal{L}^{\text{mps}}$ refines the global alignment between $\mathbf{v}^{\text{g}}_1$ and $\{\mathbf{t}^{\text{g}}_{1_1}...\mathbf{t}^{\text{g}}_{3_1}\}$.

caption per image (i.e., $K = 1$). Let $\mathbf{t}^{\text{g}}_i$ be the caption of the i-th image in the batch, and $\mathbf{v}^{\text{tc}}_{i,j}$ be the image embedding from the i-th image conditioned on the caption of image j. For the explanation in this section only, we enforce $i \neq j$. In the context of contrastive learning, image-text pairs where image, caption, and condition come from the same sample are considered positive pairs. In other words, $\langle \mathbf{v}^{\text{tc}}_{i,i}, \mathbf{t}^{\text{g}}_i \rangle$ is maximized during training, where $\langle ., . \rangle$ denotes the cosine similarity. For negative pairs, the text condition introduces multiple options. DreamLIP [60] proposed a loss that uses negatives defined as $\langle \mathbf{v}^{\text{tc}}_{i,j}, \mathbf{t}^{\text{g}}_i \rangle$ in our formulation. However, this allows to solve the contrastive objective by comparing the text condition with the text embedding, ignoring image information and creating an undesired shortcut. To overcome this problem, we instead propose to adopt $\langle \mathbf{v}^{\text{tc}}_{i,j}, \mathbf{t}^{\text{g}}_j \rangle$ as negative pairs. This ensures that image and text representations are contrasted meaningfully, i.e., neither image nor text information can be ignored. This definition generalizes to multiple captions per image (i.e., $K > 1$), where each sub-caption of the same image is considered a positive match, and negative otherwise. In this case, we can write the full notation as $\langle \mathbf{v}^{\text{tc}}_{i,i_k}, \mathbf{t}^{\text{g}}_{i_k} \rangle$ for positive pairs and $\langle \mathbf{v}^{\text{tc}}_{i,j_k}, \mathbf{t}^{\text{g}}_{j_k} \rangle$ for negative pairs, where k is the sub-caption index of the j-th image. Consequently, these positive and negative pairs allow FLAIR to learn text-aware image representations. See extended analysis in Sec. C in the supplementary.

3.3. Text-conditioned Sigmoid Loss

After constructing the positives and negatives pairs and applying $f_{\text{AttnPool}}(.)$, we adopt a contrastive loss based on the sigmoid function as proposed by SigLIP [58]. It is preferred over the InfoNCE loss [38], as it enables multiple positive pairs in the same batch, and is more efficient for multi-GPU training. Accordingly, we define our text-conditioned sigmoid loss as:

$$\mathcal{L}^{\text{tcs}}_{i,j,k} = \frac{1}{1 + e^{y_{i,j}\left(-t\langle \mathbf{v}^{\text{tc}}_{i,j_k}, \mathbf{t}^{\text{g}}_{j_k} \rangle + b\right)}} \tag{2}$$

where t is a learnable temperature, b is a learnable bias, and $\langle \cdot, \cdot \rangle$ is the cosine similarity. $y_{i,j}$ is $+1$ for positive pairs when $i = j$ for all $k \in [1, \ldots, K]$, and -1 for negative pairs otherwise. Since every batch contains B images and BK captions, we reduce the compute and memory usage of $\mathcal{L}^{\text{tcs}}_{i,j,k}$ by considering all K positive pairs, but only $B - 1$ negative pairs per image, i.e., 1 out of K captions for every negative. Therefore, we compute the similarity of $B \times (K + B - 1)$ pairs, instead of $B \times BK$ pairs for every batch.

$\mathcal{L}^{\text{tcs}}$ aligns the text-conditioned image embedding with the corresponding text embedding. Intuitively, this allows the image encoder to store semantic information locally in each token and pool the relevant tokens based on the text query producing context-aware representations. Our main experiments demonstrate that the text-conditioned image embeddings contribute significantly to fine-grained image-text alignment, providing the majority of the performance improvement in zero-shot semantic segmentation.

3.4. Multi-positive Sigmoid Loss

We find that FLAIR can be trained exclusively with the $\mathcal{L}^{\text{tcs}}$ loss. At the same time, it proves beneficial to additionally match the global image embedding $\mathbf{v}^{\text{g}}$ with every sub-caption, to also learn a coarse alignment. Following previous works [14, 29, 60], we introduce a multi-positive loss to align the global image embedding $\mathbf{v}^{\text{g}}$ with the text embedding $\mathbf{t}^{\text{g}}$ of every sub-caption. Different from previous works, we employ the contrastive sigmoid loss to han-

dle multiple positive captions per image in a more natural way. Our multi-positive sigmoid loss is defined as:

$$\mathcal{L}^{\text{mps}}_{i,j,k} = \frac{1}{1 + e^{y_{i,j}\left(-t\langle \mathbf{v}^{\text{g}}_i, \mathbf{t}^{\text{g}}_{j_k}\rangle + b\right)}} \tag{3}$$

where $y_{i,j}$ is $+1$ for all $k \in [1, \dots, K]$ positive pairs with $i = j$, and -1 for negative pairs otherwise. Equivalently to $\mathcal{L}^{\text{tcs}}$, we use all K positive pairs per image and 1 caption from every K negative sub-captions per match.

Since $\mathcal{L}^{\text{mps}}$ mainly optimizes the global image and text embeddings, it is beneficial for coarse-grained tasks. We empirically find that combining $\mathcal{L}^{\text{mps}}$ with $\mathcal{L}^{\text{tcs}}$ consistently improves performance across all tasks, particularly in zero-shot image classification, where global-level alignment is crucial. Our final loss $\mathcal{L}$ is an average of both losses and it is defined as:

$$\mathcal{L} = \frac{1}{2}(\mathcal{L}^{\text{tcs}} + \mathcal{L}^{\text{mps}}). \tag{4}$$

4. Experiments

We present our experimental evaluation of FLAIR on the three image-text retrieval settings: standard (Sec. 4.2), fine-grained (Sec. 4.3), and long (Sec. 4.4). In addition, we conduct experiments on zero-shot semantic segmentation (Sec. 4.5) and image classification (Sec. 4.6), qualitatively evaluate the attention maps of FLAIR (Sec. 4.7), and ablate important model components (Sec. 4.8).

4.1. Experimental Setup

Pre-training Datasets. To learn fine-grained image embeddings from descriptive local captions, we pre-train FLAIR on DreamLIP's [60] re-captioned datasets, which we refer to as CC3M-recap, CC12M-recap, and YFCC15M-recap. Following DreamLIP, we also merged these three datasets into a combined set of 30M samples (Merged-30M).

Implementation Details. Our model is based on the OpenCLIP [6] code implementation, adopting their default settings. We use ViT-B/16 as the vision encoder, with the default pre-processing: images are resized to 224×224 pixels, and text sequences are tokenized to a maximum length of 77 tokens. For direct comparison with DreamLIP, we follow their training configuration and caption pre-processing, splitting the MLLM-generated and original captions into individual sentences. To obtain diverse training captions, we sample $K = 8$ captions per image, with each caption randomly merging 1 to 3 sentences (i.e., $S = 3$). To maximize sampling variability while retaining context, we randomly construct our sub-caption by either sampling consecutive sentences or merging sentences from random positions in the original text. For fair comparison, we reproduce CLIP [41] and SigLIP [58] on all re-captioned datasets under identical training configurations as FLAIR. Further details are available in Sec. E in the supplementary.

Inference with FLAIR. To utilize the fine-grained embeddings from FLAIR for image-to-text (I2T) retrieval, each image i is first conditioned on all j texts to generate the conditioned embeddings $\mathbf{v}^{\text{tc}}_{i,j}$. Then we compute the similarity scores between the conditioned embeddings and each text embedding ($\langle \mathbf{v}^{\text{tc}}_{i,j}, \mathbf{t}^{\text{g}}_j \rangle$) to obtain Recall@K from the top-K retrieval items. The text-to-image (T2I) retrieval score matrix is the transpose of the image-to-text retrieval matrix. The inference can be further sped-up by the top-K selection approach in Sec. E in the supplementary.

4.2. Standard Zero-shot Image-text Retrieval

As a standard assessment of image-text alignment, we follow prior works [26, 54, 60] to evaluate image-text retrieval on the validation splits of MSCOCO [30] and Flickr30K [40], where each image is typically paired with five global captions.

Results. We report the results on the standard retrieval task in the left side of Tab. 1. FLAIR outperforms the three baselines, CLIP, SigLIP, and DreamLIP on all pre-training datasets by a large margin. Comparing models trained on CC3M-recap, FLAIR surpasses DreamLIP in the retrieval task, obtaining higher R@1 scores on both COCO (T2I: +7.9%, I2T: +10.8%) and Flickr30k (T2I: +12.1%, I2T: +9.5%) datasets. When including SOTA models, FLAIR trained on CC12M-recap obtains a similar performance to SigLIP trained on 10B samples, and surpasses it significantly once we move to larger datasets such as YFCC15M-recap and the merged 30M samples. FLAIR-30M (vs. SigLIP-10B) achieves 81.1% (vs. 75.6%) T2I, 94.7% (vs. 89.1%) I2T on Flickr30k and is similarly better on MSCOCO. We also notice that CLIP and SigLIP trained on YFCC15M-recap can match or surpass their counterparts trained on billions of data samples. This suggests two key insights: 1) text augmentations from long synthetic captions empowers VLMs with better retrieval capability, and 2) FLAIR with text-conditioned attention pooling generates more targeted image embeddings for retrieval, maximizing the benefits from text augmentations, and resulting in a significant improvement with much less image data.

4.3. Fine-grained Zero-shot Image-text Retrieval

Standard retrieval tasks do not fully capture a model's ability to align detailed descriptions with images. To address this, we introduce a fine-grained retrieval task aimed at evaluating how well a model can associate an image with fine-grained captions. Our benchmark is constructed as follows: 1) We use the recently released densely-captioned datasets DOCCI [37] and IIW [19]. Due to the careful human annotation process, their long captions are free from hallucinations; 2) For each test image in DOCCI and IIW, we split the long captions into individual sentences, yielding an average of 7.1 captions per image in DOCCI and 10.1 in IIW. As

Setting	Method	Standard Retrieval								Fine-grained Retrieval							
		MSCOCO				Flickr30k				DOCCI-FG				IIW-FG			
		T2I		I2T		T2I		I2T		T2I		I2T		T2I		I2T	
		R@1	R@5	R@1	R@5	R@1	R@5	R@1	R@5	R@1	R@5	R@1	R@5	R@1	R@5	R@1	R@5
CC3M-recap	CLIP [41]	27.0	52.6	38.9	66.1	49.9	77.0	67.8	88.5	10.3	23.4	25.0	50.9	24.4	45.0	61.1	85.6
	SigLIP [58]	28.3	54.4	40.1	67.5	53.2	78.5	69.9	90.4	10.4	23.8	24.9	50.6	24.7	45.1	62.6	86.9
	DreamLIP [60]	29.8	55.4	40.8	68.4	53.6	78.4	69.2	91.5	10.3	22.8	23.3	47.5	22.7	41.6	59.2	83.3
	FLAIR	**37.7**	**64.5**	**51.6**	**77.2**	**65.7**	**86.8**	**78.7**	**95.2**	**15.1**	**30.9**	**35.7**	**63.5**	**30.5**	**52.3**	**70.6**	**90.9**
CC12M-recap	CLIP [41]	39.8	66.4	56.2	80.5	67.0	87.7	81.7	96.5	16.0	31.6	39.5	66.5	31.8	52.9	76.4	93.6
	SigLIP [58]	40.4	67.0	55.3	79.7	66.7	88.1	82.5	96.1	16.2	31.9	40.0	66.8	31.9	53.2	78.4	94.3
	DreamLIP [60]	40.6	66.5	54.0	78.3	68.3	89.3	84.1	97.8	17.2	33.0	41.6	68.5	31.9	52.1	77.8	94.9
	FLAIR	**47.8**	**73.5**	**64.1**	**85.0**	**75.4**	**92.15**	**90.8**	**98.4**	**21.4**	**38.8**	**50.4**	**76.7**	**38.7**	**59.9**	**83.8**	**96.9**
YFCC15M-recap	CLIP [41]	44.7	71.2	61.0	85.0	72.3	90.8	89.1	97.6	18.1	35.3	43.1	71.9	34.4	56.5	81.4	96.7
	SigLIP [58]	46.6	72.8	62.6	85.3	73.6	92.1	90.0	97.6	18.9	35.8	46.3	74.8	35.5	56.6	84.3	96.2
	DreamLIP [60]	42.4	68.5	57.0	81.0	70.0	89.2	87.3	98.1	17.3	33.6	41.4	69.8	32.0	53.0	76.1	95.4
	FLAIR	**51.2**	**76.0**	**67.3**	**88.1**	**79.2**	**94.2**	**93.3**	**99.1**	**23.0**	**41.2**	**53.7**	**79.7**	**39.5**	**62.1**	**85.5**	**96.4**
SOTA Comparison	OpenCLIP (2B) [6]	41.7	67.1	59.3	82.4	71.9	90.4	87.5	97.7	17.4	31.9	49.7	75.9	30.6	48.4	84.1	95.4
	SigLIP (10B) [58]	47.2	72.1	65.5	86.2	75.6	92.8	89.1	98.6	20.6	35.9	57.5	82.1	33.8	53.0	83.7	97.7
	MetaCLIP (2.5B) [53]	41.4	67.2	59.4	80.6	76.2	90.7	85.9	97.3	-	-	-	-	-	-	-	-
	Llip (2.5B) [26]	45.6	70.8	63.4	84.3	75.1	92.8	90.1	98.5	-	-	-	-	-	-	-	-
	DreamLIP (30M) [60]	44.8	69.8	62.3	84.5	73.3	91.8	89.9	99.0	21.6	39.3	51.2	78.3	37.5	58.6	85.3	97.4
	FLAIR (30M)	**53.3**	**77.5**	**68.0**	**87.8**	**81.1**	**94.9**	**94.7**	**99.3**	**25.0**	**43.8**	**59.0**	**84.1**	**41.7**	**63.4**	**91.5**	**98.9**

Table 1. Zero-shot image-text retrieval on validation splits for standard benchmarks (Flickr30k [40] and MSCOCO [30]) and our introduced fine-grained retrieval setting (sentence-level on DOCCI [37] and IIW [19]). Except for "SOTA Comparison", all models are pre-trained on CC3M-recap, CC12M-recap, YFCC15M-recap, under the same training configurations. All models use ViT-B/16 as the vision encoder.

Method	Data	DCI		SV-1k		SV-10k		Urban-1k	
		I2T	T2I	I2T	T2I	I2T	T2I	I2T	T2I
OpenCLIP [41]	2B	56.0	55.4	90.3	87.7	69.6	66.8	69.5	65.8
LiT [57]	100M	41.7	40.9	86.0	80.0	61.4	50.6	-	-
ALIGN [21]	700M	56.5	57.4	86.3	85.3	65.1	62.7	-	-
SigLIP [58]	10B	57.7	56.2	85.8	83.4	83.4	63.0	62.7	62.1
Long-CLIP [59]	400M	47.4	44.1	90.6	87.4	73.1	62.0	78.9	79.5
LoTLIP [51]	100M	**62.1**	61.0	95.5	86.8	81.4	83.7	-	-
FLAIR	3M	47.3	50.5	91.0	89.7	72.0	70.6	63.5	69.5
FLAIR	12M	55.5	60.8	96.1	95.1	85.0	83.4	74.6	80.6
FLAIR	15M	54.9	62.4	97.4	96.7	88.8	86.8	82.4	86.6
FLAIR	30M	61.3	**66.2**	**98.5**	**98.0**	**90.3**	**89.4**	**83.6**	**87.7**

Table 2. Zero-shot long text-image retrieval tasks. I2T and T2I indicate the R@1 score on image-to-text and text-to-image retrieval, respectively. The best results are **bold**, second-best are underlined. All models use ViT-B/16 as vision encoder.

shown in Fig. 10, each caption focuses on a specific local part of the image, making both T2I and I2T tasks significantly more challenging than standard retrieval. We refer to our split datasets as DOCCI-FG and IIW-FG.

Results. The results obtained on DOCCI-FG and IIW-FG are reported in the right side of Tab. 1. The difficulty of this task is evident from the significantly lower text-to-image (T2I) retrieval scores compared to standard retrieval. Despite that, FLAIR consistently outperforms baselines across all training configurations and even surpasses the CLIP and SigLIP models trained on billions of samples. Interestingly, FLAIR trained on CC12M-recap achieves higher R@1 scores (38.7%), in terms of T2I retrieval on IIW-FG, compared to SigLIP-10B (33.8%).

On the 30M dataset, the performance of FLAIR further improves to 41.7%, outperforming DreamLIP by 4.2% in T2I retrieval (R@1). Overall, FLAIR achieves an increased between 3.4% and 7.8% in R@1 scores compared to DreamLIP. These results demonstrate that FLAIR learns to align images with detailed, fine-grained captions more effectively than the baselines.

4.4. Long Zero-shot Image-Text Retrieval

Image-text retrieval with long captions imposes a unique challenge for CLIP models. Following LoTLIP [51] and Long-CLIP [59], we evaluate FLAIR on datasets with long captions, including DCI [49], 1k (SV-1k) and 10k (SV-10k) subsets of ShareGPT-4V [5], and Urban-1k [59], with the results presented in Tab. 2.

Unlike previous methods specifically designed for long-caption retrieval with extended token limits and larger text encoders, FLAIR employs the standard CLIP text encoder

Method	Data Size	VOC20	Cityscapes	Context59	ADE20K	COCO-Stuff	Average
CLIP [41]	400M	41.8	5.5	9.2	3.2	4.4	12.8
OpenCLIP [6]	2B	47.2	5.1	9.0	2.9	5.0	13.9
MetaCLIP [53]	2.5B	35.4	5.0	8.1	2.2	4.3	11.0
CLIP [58]		11.3	5.0	4.5	1.3	2.8	5.0
SigLIP [58]	30M	14.5	5.5	5.8	2.2	3.8	6.4
DreamLIP [60]		1.8	0.9	0.4	0.1	0.1	0.7
FLAIR	3M	60.9	**20.6**	**23.8**	13.2	13.1	26.3
FLAIR	12M	69.7	20.1	22.9	**13.3**	**15.4**	**28.3**
FLAIR	15M	66.7	16.5	17.4	9.1	13.6	24.7
FLAIR	30M	**73.0**	13.6	18.6	10.4	13.3	25.8

Table 3. Mean intersection over union (mIoU) for zero-shot semantic segmentation on the VOC20 [13], Cityscapes [8], Context59 [34], ADE20K [61], and COCO-Stuff [2] datasets. All models employ ViT-B/16 as vision encoder.

Method	Data Size	Food-101	CIFAR-10	CIFAR-100	SUN397	Cars	Aircraft	DTD	Pets	Caltech-101	Flowers	ImageNet	Average
LaCLIP [14]	3M	14.2	57.1	27.5	35.1	1.6	1.6	16.6	15.6	52.7	14.7	21.5	23.5
MLLM-A [31]		18.7	58.4	32.4	43.8	3.9	1.5	20.2	32.1	63.5	17.5	25.0	28.8
CLIP [41]		17.9	75.0	40.8	43.1	2.6	1.0	15.3	22.1	68.9	12.6	23.8	29.4
SigLIP [58]		18.4	76.4	41.9	46.9	3.0	1.4	17.6	20.6	70.4	10.8	25.4	30.3
DreamLIP [60]		23.1	75.9	44.2	46.6	3.4	1.6	19.0	27.4	66.1	16.0	30.1	32.1
FLAIR		**24.2**	**82.0**	**51.5**	**53.8**	**3.7**	**1.7**	**23.9**	**34.2**	**70.1**	**19.1**	**33.8**	**36.2**
CLIP [41]	30M	61.3	92.2	66.9	62.2	19.3	5.7	30.9	49.3	83.7	43.4	50.0	51.4
SigLIP [58]		64.2	91.0	67.6	64.0	22.0	5.7	33.5	53.3	84.3	43.6	51.0	52.7
DreamLIP [60]		**75.4**	92.3	**70.7**	63.7	22.7	**7.9**	33.9	**64.1**	**88.0**	**51.1**	**58.1**	**57.0**
FLAIR		72.5	**93.1**	69.6	**66.9**	**31.1**	7.2	**37.3**	55.6	86.5	48.4	56.6	56.8
OpenCLIP [6]	2B	86.2	94.8	76.5	70.0	87.4	25.8	54.9	89.5	93.2	69.8	70.2	74.4
MetaCLIP [53]	2.5B	88.3	**95.7**	79.0	68.5	82.9	30.3	62.1	**91.7**	93.3	73.9	72.1	76.2
Llip [26]	2.5B	**89.0**	**95.7**	**81.4**	**70.9**	**88.2**	**41.5**	**63.7**	**93.5**	94.7	**74.9**	**75.3**	**79.0**

Table 4. Top-1 accuracy for zero-shot classification on: Food-101 [1], CIFAR-10 & CIFAR-100 [24], SUN397 [52], Cars [23], Aircraft [33], DTD [7], Pets [39], Caltech-101 [15], Flowers [36], ImageNet [10]. All models use ViT-B/16 as vision encoder. The best and second-best results are **bold** and underlined.

with a 77-token limit. The former SOTA, LoTLIP [51], was trained on a 100M-scale re-captioned dataset, while LongCLIP [59] fine-tunes a 400M-scale CLIP model with an additional 1M images with long captions. Although FLAIR is trained on a smaller training set of 30M samples, it still outperforms these methods on SV-1k, SV-10k, and Urban-1k. Most notably on T2I, FLAIR obtained improvements of 10.4%, 5.7%, and 8.2% in terms of R@1 over the previous SOTA. Remarkably, FLAIR trained on 15M samples already surpasses all previous methods on 3 out of 4 datasets.

This significant performance gain can be explained as follows: 1) text-conditioned attention pooling can adapt to the rich semantics in long texts to extract all relevant visual information. 2) By sampling diverse captions the model becomes aligned with the distribution of long captions.

4.5. Zero-shot Semantic Segmentation

For VLMs, zero-shot semantic segmentation involves measuring the similarity $\{\langle \mathbf{v}_i^{\text{loc}}, \mathbf{t}_j^{\text{g}} \rangle \mid j \in \{1, 2, \ldots, M\}\}$ for M different class names. Recent works [12, 25, 50] provide a framework to map these similarities to semantic segmentation outputs. To examine the raw alignment of local image tokens $\mathbf{v}^{\text{loc}}$ with the corresponding input texts, we perform semantic segmentation following [50] without post-processing or segmentation-specific modifications.

As shown in Tab. 3, FLAIR trained on all subsets of data, consistently outperforms CLIP-based methods trained on significantly larger datasets, achieving an improvement of 10.1% - 25.8% mIoU increase across all datasets (14.4% on average). As illustrated in Fig. 1, DreamLIP's $\mathbf{v}^{\text{loc}}$ image token embeddings show weak correspondence to the input text, which we conclude is the result of their choice of negatives as discussed in Sec. 3.2. In contrast, FLAIR, optimized with $\mathcal{L}^{\text{tcs}}$, effectively aligns $\mathbf{v}^{\text{loc}}$ with varying text prompts, demonstrating strong localization capabilities.

4.6. Zero-shot Image Classification

Following [14, 60], we evaluate the zero-shot classification performance of FLAIR and baseline methods on ImageNet [10] and 10 additional datasets, as shown in Tab. 4. In retrieval tasks, text-augmented methods outperform VLMs trained on billions of images. However, in image classification they lag behind by around 20%. This demonstrates that scaling up the number of images remains a key factor in improving VLM's classification performance.

When trained on CC3M-recap, FLAIR achieves a 4.1% higher average performance than DreamLIP and other baselines. This shows that FLAIR, although optimized to generate fine-grained visual representations, could still efficiently gain global-level visual understanding performance when images are relatively scarce. However, when scaled up to 30M samples, FLAIR, while still outperforming CLIP and SigLIP by 4%, is on par with DreamLIP (-0.2% on avg.). This shows that these methods trained on synthetic captions converge similarly, further suggesting the importance of scaling up images for the classification task. Therefore, we hypothesize that scaling FLAIR to larger datasets would extend the concept vocabulary and image coverage, closing the gap to large-scale models on zero-shot classification.



Figure 4. Visualization of attention maps in the attention pooling layer $f_{\text{AttnPool}}(.)$. Regions of high attention are highlighted in red.

4.7. Attention Maps Visualization

For a given image with two different local captions, we visualize the attention maps of $f_{\text{AttnPool}}(.)$, i.e., which image tokens are pooled together, in Fig. 4. As illustrated by the "truck" and "worker" example, FLAIR can locate both large and small objects in an image. The horses example shows that FLAIR is able to differentiate objects by their individual properties such as color and location. Notably, FLAIR is also precise in identifying individual parts of an object, exemplified by the eyes, mouth, and paw of the dog, where the focus lies on the one that is raised. These results show FLAIR's strong sensitivity to semantic details. More visualizations can be found in Sec. A in the supplementary.

4.8. Ablation Study

Model Components. In Tab. 5, we analyze the components of FLAIR: text conditioning (TC), global loss (GL), multiple captions per image (MC), and diverse caption sampling instead of single sentences (DS). $\mathcal{L}^{\text{tcs}}$ and $\mathcal{L}^{\text{mps}}$ correspond to TC+MC and GL+MC respectively.

SigLIP is equivalent to only using GL (1). Replacing GL with TC (2) leads to performance improvements across all metrics, achieving a 3.7%/5.5% increase in R@1 for MSCOCO retrieval and a 33.8% boost in VOC20 segmentation demonstrating its contribution to the fine-grained alignment. Adding MC improves performance in both scenarios (3 and 4). Our diverse sampling (DS) is another significant improvement, especially in segmentation and long-retrieval on the Urban-1K which gains over 20% (5 and 6). Combining all with FLAIR (7) achieves the best performance in all but long-retrieval. In summary, $\mathcal{L}^{\text{tcs}}$ is foundational to our method's performance, sampling diverse captions provides a substantial boost in long retrieval tasks, while $\mathcal{L}^{\text{mps}}$ delivers additional gains, particularly for global-level tasks.

	Method				COCO		DOCCI		Urban-1K		VOC20	ImageNet
	GL	TC	MC	DS	T2I	I2T	T2I	I2T	T2I	I2T	mIOU	Top-1
1	✓				28.3	40.1	10.4	24.9	42.8	40.5	3.1	25.4
2		✓			32.0	45.6	12.7	30.9	44.4	42.6	36.9	28.1
3	✓		✓		32.9	44.6	13.3	31.0	47.9	46.5	1.7	27.9
4		✓	✓		34.8	47.1	14.1	30.2	46.6	40.9	34.1	29.4
5	✓		✓	✓	35.0	49.1	13.0	33.1	**70.7**	**64.6**	7.2	32.0
6		✓	✓	✓	36.2	50.0	13.8	34.6	68.3	63.1	46.5	31.5
7	✓	✓	✓	✓	**37.7**	**51.6**	**15.1**	**35.7**	69.5	63.5	**59.7**	**33.8**

Table 5. Ablation study on different components of FLAIR on the CC3M-recap dataset. **GL**: Global Loss, **TC**: Text Conditioning, **MC**: Multiple Captions, **DS**: Diverse Sampling.

Additional Ablations. In supplementary Sec. D, we provide additional ablations to support important choices. We pre-trained FLAIR on the original CC3M dataset and on the PixelProse [46] dataset with synthetic captions generated by Gemini-Pro [42], showing that our method is not restricted to long captions and adaptable to a variety of data distributions. By varying the sampling strategy of the diverse captions, we find that it is crucial across tasks to sample both short and long captions instead of only a fixed length. By testing a different number of multiple captions K ranging from 2 to 10, we observe that performance converges at around 8 captions. Finally, we ablate the maximal number of sampled sentences S and observe that merging 3 sentences achieved the most balanced results.

5. Conclusion and Limitations

We introduce FLAIR, a VLM that learns **F**ine-grained **La**nguage-informed **I**mage **R**epresentations by conditioning on the semantics in dense local captions. Trained on 30M recaptioned images, FLAIR outperforms baselines trained on billions of images across standard, fine-grained, and long-form image-text retrieval tasks. The significant improvements in zero-shot segmentation compared to the baselines as well as the qualitative results corroborate that FLAIR learns a fine-grained alignment between text and image at the token-level.

While FLAIR matches baselines trained on the same number of images in zero-shot classification, it still falls behind CLIP models trained on significantly larger datasets. This suggests that, although leveraging detailed synthetic captions enhances fine-grained image understanding, it does not replace the image coverage and conceptual richness of larger datasets for global-level tasks. To tackle this limitation, a natural future direction involves scaling the synthetic recaptioning to large-scale datasets and training variants of FLAIR with higher parameter count.

Acknowledgments This work was partially funded by the ERC (853489 - DEXIM), the German Federal Ministry of Education and Research (BMBF, grant number: 01IS18039A), and the Alfried Krupp von Bohlen und Halbach Foundation, which we thank for their generous support. The authors gratefully acknowledge the Gauss Centre for Supercomputing e.V. (www.gauss-centre.eu) for funding this project by providing computing time on the GCS Supercomputer JUWELS [22] at Jülich Supercomputing Centre (JSC).

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